

Lecture 7: Journaling vs. LFS and Performance Evaluation

- This lecture combines design comparison with quantitative performance evaluation.
- The objective is not merely to list features, but to select metrics and analyze behavior for a workload.

Learning Objectives

- Distinguish journaling from log-structured storage.
- Define throughput, latency and IOPS.
- Explain workload characteristics affecting file-system performance.
- Design a simple file-system benchmark.
- Interpret performance results and trade-offs.

Journaling vs. LFS: Core Idea

	Journaling	LFS
Primary purpose	Crash consistency/recovery	Make updates sequential/log-like
Main write idea	Record changes in journal, then apply	Append new versions to log
Main challenge	Extra journal I/O	Cleaning/write amplification
Recovery	Journal replay	Checkpoint/log recovery

- **Throughput:** amount of useful data processed per unit time.

$$T = \frac{\text{bytes transferred}}{\text{elapsed time}}$$

- **Latency:** time required to complete an operation.
- **IOPS:** I/O operations completed per second.
- **CPU utilization:** processor cost of file-system activity.
- **Recovery time:** time required to return to a usable consistent state.

Workload Characterization

Sequential workload

- Large reads/writes
- Streaming data
- Backup/archive

Random workload

- Small reads/writes
- Metadata-heavy activity
- Database-style access

The same file-system design can behave very differently under these workloads.

Factors Affecting Performance

- Block size and allocation granularity
- Cache/buffer size
- Read-ahead and write buffering
- Sequential vs. random I/O
- Fragmentation
- Metadata overhead
- Journaling mode
- Storage technology: HDD vs. SSD/NVMe
- Queue depth and concurrency

Benchmark Design

- 1 Define the workload.
- 2 Define the file-system configuration.
- 3 Select measurable metrics.
- 4 Run a controlled benchmark.
- 5 Repeat measurements to reduce noise.
- 6 Compare results and explain the causes.

Important: change one major variable at a time where possible.

Example Evaluation

Scenario

A server performs 20,000 small random writes per second and must recover quickly after power failure.

Ask:

- Which metrics matter most?
- What role does journaling play?
- What storage technology is being used?
- How would write amplification affect endurance/performance?
- What experiment would validate the design?

Journaling vs. LFS: Trade-Off Analysis

Criterion	Journaling	LFS
Crash consistency	Strong focus	Designed with recovery support
Sequential writes	Journal is sequential, main FS may not be	Core design principle
Cleaning	Usually not LFS-style segment cleaning	Required
Extra writes	Journal traffic	Cleaning + metadata traffic
Best fit	Conventional FS needing consistency	Workloads benefiting from log-style writes

- **Galvin, 10th Ed.:** Ch. 14 – Efficiency and Performance; Ch. 15 – Log-Structured File Systems and Recovery.
- **Stallings, 9th Ed.:** Ch. 11 – I/O management, disk scheduling and caching; Ch. 18 – File Management and secondary-storage management.

- Performance must be evaluated relative to workload.
- Throughput, latency, IOPS, CPU overhead and recovery time are complementary metrics.
- Journaling and LFS solve related but different design problems.
- A defensible design decision requires measurable evidence, not only feature comparison.